Dynamic Social Optimum: 3×3 Grid Network

COMP3821

1 Definition and reduction to path assignments

The simulator uses linear edge costs $c_e(f) = a_e + b_e f$ and the total social cost

$$SC = \sum_t \sum_{e \in E} f_e(t)\, c_e\big(f_e(t)\big),$$

where $f_e(t)$ is the number of agents that traverse e during timestep t. The *dynamic social optimum* is the lowest value of this expression over every feasible way of routing the agents:

$$\boxed{SC_{\text{opt}} = \min_{\text{feasible routing decisions}} \sum_t \sum_{e \in E} f_e(t)\, c_e\big(f_e(t)\big)}$$

Feasible routings. The movement rules of the specification (Section 3.1) and the simulator fix what "feasible" means:

1. all N agents are released at $t = 0$ (none of these scenarios uses cascades);
2. at every timestep each unfinished agent traverses exactly one edge, so no agent can wait;
3. with `forward_only` (the default), an agent never moves to a node with smaller hop distance from its origin and never revisits a node.

Lemma 1 (Routing = path assignment)**.** *Let $\mathcal{P}_{od}$ be the simple paths from o to d that obey rule 3. Every feasible routing gives each agent a path $P_i \in \mathcal{P}_{o_i d_i}$, and agent i traverses the k-th edge of P_i at timestep $t = k - 1$. Every assignment of such paths can be realised. Therefore*

$$SC = \sum_{e \in E} \sum_{t \geq 0} X_{e,t}\, \big(a_e + b_e X_{e,t}\big), \qquad X_{e,t} = \#\{\, i : e \text{ is the } (t+1)\text{-th edge of } P_i \,\},$$

and SC_{opt} is the minimum of this over all path assignments.

Proof. Rule 2 means an agent's position at time t is the end of the t-th edge it has taken. Rule 3 means the edges it takes form a simple path in the allowed subgraph. So $f_e(t) = X_{e,t}$. Conversely, a central planner can tell each agent which next edge to commit to at each step. The outcome depends only on the paths, so every path assignment is achievable. □

So the dynamic problem is a *static* problem on the *time-expanded* resources (e, t). Agents on the same edge only slow each other down if they cross it in the same step.

Lemma 2 (When the dynamic and static optima coincide)**.** *Suppose every edge e sits at the same position $k(e)$ on every feasible path that contains it. Then $X_{e,t} = x_e\, \mathbf{1}[t = k(e) - 1]$, where x_e is the total number of agents using e. So $SC = \sum_e x_e\, c_e(x_e)$, which is the classical static social-optimum objective (Beckmann; Roughgarden) restricted to integer flows.*

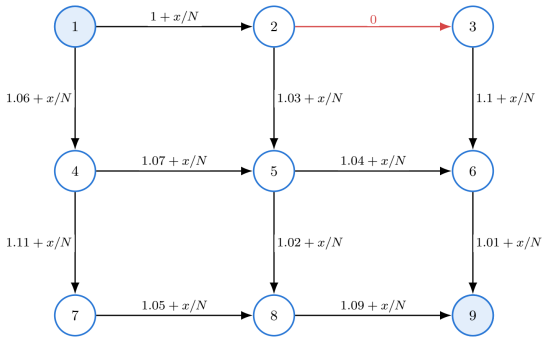
Integer versus continuous. Agents are indivisible, so SC_{opt} is an integer program. Relaxing the path counts to non-negative reals gives a convex quadratic program. Its minimum is a lower bound on SC_{opt} and equals the limit of SC_{opt}/N for large N. Both are reported below. Because c_e is linear, the cost one extra agent adds to resource (e, t) is

$$\Delta^{+}_{e,t} = (X+1)\,c_e(X+1) - X\,c_e(X) = a_e + b_e(2X+1), \qquad X = X_{e,t},$$

and in the continuous relaxation the marginal cost is $\frac{d}{dX}\big[X c_e(X)\big] = a_e + 2b_e X$.

Scaling with N. Every small scenario sets $b_e = \beta_e/N$ with β_e fixed, so the per-agent optimum SC_{opt}/N is almost the same for $N = 20, 50, 80$.

2 Network



Nodes $1, \ldots, 9$ form a 3×3 grid in row-major order. Edges point right or down, and all agents travel $1 \to 9$. Every edge has $b_e = 1/N$ (0.05, 0.02 and 0.0125 for $N = 20, 50, 80$), except $2 \to 3$, which is free ($a = b = 0$).

Edge	a_e	Edge	a_e	Edge	a_e
$1 \to 2$	1.00	$1 \to 4$	1.06	$2 \to 3$	0
$4 \to 5$	1.07	$4 \to 7$	1.11	$2 \to 5$	1.03
$5 \to 6$	1.04	$5 \to 8$	1.02	$3 \to 6$	1.10
$7 \to 8$	1.05	$8 \to 9$	1.09	$6 \to 9$	1.01

There are $\binom{4}{2} = 6$ monotone paths, written as R(ight)/D(own) strings:

Name	Path	$\sum a_e$
RRDD	$1 \to 2 \to 3 \to 6 \to 9$	3.11
RDRD	$1 \to 2 \to 5 \to 6 \to 9$	4.08
RDDR	$1 \to 2 \to 5 \to 8 \to 9$	4.14
DRRD	$1 \to 4 \to 5 \to 6 \to 9$	4.18
DRDR	$1 \to 4 \to 5 \to 8 \to 9$	4.24
DDRR	$1 \to 4 \to 7 \to 8 \to 9$	4.31

3 Derivation

Timing. The node in row r, column c (0-indexed) is reached after exactly $r + c$ steps on every path, so an edge leaving it is always crossed at $t = r + c$. Lemma 2 applies, and $SC = \sum_e x_e(a_e + b_e x_e)$ with x_e the edge flows. The dynamic optimum equals the static one.

Continuous optimum (KKT with node potentials). Write $y_e = x_e/N$. Then $SC/N = \sum_e (a_e y_e + y_e^2)$, summed over the 11 congestible edges ($2 \to 3$ costs nothing). This is a convex min-cost flow problem with marginal cost $m_e = a_e + 2y_e$ on every edge except $m_{23} = 0$. A flow y is optimal exactly when there are node potentials π with

$$\pi_v - \pi_u \le m_e(y_e) \text{ for every edge } (u,v), \qquad \text{with equality whenever } y_e > 0.$$

Here π_v is the marginal cost of reaching v. We guess that every edge is used and solve the linear system

$$\pi_v - \pi_u = a_e + 2y_e \ \ (\text{every } e \ne 2 \to 3), \qquad \pi_3 = \pi_2, \qquad \text{flow conservation}, \quad \pi_1 = 0,$$

which has 20 equations and 20 unknowns. The unique solution (exact fractions) is:

Edge	y_e		Edge	y_e	
$1 \to 2$	2061/3400	0.60618	$1 \to 4$	1339/3400	0.39382
$2 \to 3$	959/1700	0.56412	$2 \to 5$	143/3400	0.04206
$3 \to 6$	959/1700	0.56412	$4 \to 5$	139/680	0.20441
$4 \to 7$	161/850	0.18941	$5 \to 6$	63/1700	0.03706
$5 \to 8$	89/425	0.20941	$7 \to 8$	161/850	0.18941
$6 \to 9$	511/850	0.60118	$8 \to 9$	339/850	0.39882

$$\pi = (\pi_1, \dots, \pi_9) = \big(0,\ 2.21235,\ 2.21235,\ 1.84765,\ 3.32647,\ 4.44059,\ 3.33647,\ 4.76529,\ 6.65294\big)$$

Every $y_e > 0$, so the guess was consistent. All KKT conditions hold with equality, and since the objective is convex they are sufficient. Hence

$$\frac{SC^*}{N} = \sum_e \big(a_e y_e + y_e^2\big) = \frac{872297}{170000} \approx 5.131159.$$

The edge flows are unique, but their split into paths is not. One decomposition is RRDD 0.56412, RDRD 0.03637, RDDR 0.00568, DRRD 0.00068, DRDR 0.20373, DDRR 0.18941. More than half the agents take RRDD, the free $2 \to 3$ edge, and the rest spread out to avoid congesting it.

Integer optimum (exact, with certificate). For integer flows, the condition becomes a discrete one. An integer flow x is optimal if and only if the residual network, with forward cost $\Delta_e^+ = a_e + b_e(2x_e + 1)$ and backward cost $-\Delta_e^- = -(a_e + b_e(2x_e - 1))$ (only when $x_e > 0$), has no negative cycle. Equivalently, there are potentials π with

$$a_e + b_e(2x_e - 1) \ \le\ \pi_v - \pi_u \ \le\ a_e + b_e(2x_e + 1) \qquad (\text{lower bound only if } x_e > 0).$$

No rerouting of any set of agents can then lower the cost. Table 1 gives the optimal integer routings. Tables 2–4 list the potentials and check every inequality.

Path	$N = 20$	$N = 50$	$N = 80$
RRDD (1→2→3→6→9)	11	28	45
RDRD (1→2→5→6→9)	1	2	3
RDDR (1→2→5→8→9)	0	0	1
DRRD (1→4→5→6→9)	0	0	0
DRDR (1→4→5→8→9)	4	10	16
DDRR (1→4→7→8→9)	4	10	15
SC_{opt}	102.64	256.58	410.5075
SC_{opt}/N	5.13200	5.13160	5.13134

Table 1: An optimal integer routing for each population size (exact).

Edge $e = (u, v)$	a_e	x_e	$a_e + b_e(2x_e - 1)$	$\pi_v - \pi_u$	$a_e + b_e(2x_e + 1)$
$1 \to 2$	1	12	2.1500	2.2500	2.2500
$1 \to 4$	1.06	8	1.8100	1.9100	1.9100
$2 \to 3$	0	11	0.0000	0.0000	0.0000
$2 \to 5$	1.03	1	1.0800	1.1600	1.1800
$3 \to 6$	1.1	11	2.1500	2.2500	2.2500
$4 \to 5$	1.07	4	1.4200	1.5000	1.5200
$4 \to 7$	1.11	4	1.4600	1.5600	1.5600
$5 \to 6$	1.04	1	1.0900	1.0900	1.1900
$5 \to 8$	1.02	4	1.3700	1.4700	1.4700
$6 \to 9$	1.01	12	2.1600	2.2600	2.2600
$7 \to 8$	1.05	4	1.4000	1.4100	1.5000
$8 \to 9$	1.09	8	1.8400	1.8800	1.9400

Table 2: Optimality certificate for $N = 20$ ($b_e = 1/20$). Potentials: $\pi_1 = 0$, $\pi_2 = 2.2500$, $\pi_3 = 2.2500$, $\pi_4 = 1.9100$, $\pi_5 = 3.4100$, $\pi_6 = 4.5000$, $\pi_7 = 3.4700$, $\pi_8 = 4.8800$, $\pi_9 = 6.7600$

Edge $e = (u, v)$	a_e	x_e	$a_e + b_e(2x_e - 1)$	$\pi_v - \pi_u$	$a_e + b_e(2x_e + 1)$
$1 \to 2$	1	30	2.1800	2.2200	2.2200
$1 \to 4$	1.06	20	1.8400	1.8700	1.8800
$2 \to 3$	0	28	0.0000	0.0000	0.0000
$2 \to 5$	1.03	2	1.0900	1.1300	1.1300
$3 \to 6$	1.1	28	2.2000	2.2400	2.2400
$4 \to 5$	1.07	10	1.4500	1.4800	1.4900
$4 \to 7$	1.11	10	1.4900	1.4900	1.5300
$5 \to 6$	1.04	2	1.1000	1.1100	1.1400
$5 \to 8$	1.02	10	1.4000	1.4400	1.4400
$6 \to 9$	1.01	30	2.1900	2.2300	2.2300
$7 \to 8$	1.05	10	1.4300	1.4300	1.4700
$8 \to 9$	1.09	20	1.8700	1.9000	1.9100

Table 3: Optimality certificate for $N = 50$ ($b_e = 1/50$). Potentials: $\pi_1 = 0$, $\pi_2 = 2.2200$, $\pi_3 = 2.2200$, $\pi_4 = 1.8700$, $\pi_5 = 3.3500$, $\pi_6 = 4.4600$, $\pi_7 = 3.3600$, $\pi_8 = 4.7900$, $\pi_9 = 6.6900$

Edge $e = (u, v)$	a_e	x_e	$a_e + b_e(2x_e - 1)$	$\pi_v - \pi_u$	$a_e + b_e(2x_e + 1)$
$1 \to 2$	1	49	2.2125	2.2125	2.2375
$1 \to 4$	1.06	31	1.8225	1.8475	1.8475
$2 \to 3$	0	45	0.0000	0.0000	0.0000
$2 \to 5$	1.03	4	1.1175	1.1175	1.1425
$3 \to 6$	1.1	45	2.2125	2.2375	2.2375
$4 \to 5$	1.07	16	1.4575	1.4825	1.4825
$4 \to 7$	1.11	15	1.4725	1.4975	1.4975
$5 \to 6$	1.04	3	1.1025	1.1200	1.1275
$5 \to 8$	1.02	17	1.4325	1.4525	1.4575
$6 \to 9$	1.01	48	2.1975	2.2225	2.2225
$7 \to 8$	1.05	15	1.4125	1.4375	1.4375
$8 \to 9$	1.09	32	1.8775	1.8900	1.9025

Table 4: Optimality certificate for $N = 80$ ($b_e = 1/80$). Potentials: $\pi_1 = 0$, $\pi_2 = 2.2125$, $\pi_3 = 2.2125$, $\pi_4 = 1.8475$, $\pi_5 = 3.3300$, $\pi_6 = 4.4500$, $\pi_7 = 3.3450$, $\pi_8 = 4.7825$, $\pi_9 = 6.6725$

4 Result

N	SC_{opt}	SC_{opt}/N	Continuous bound/N
20	102.6400	5.13200	5.13116
50	256.5800	5.13160	5.13116
80	410.5075	5.13134	5.13116

$$\boxed{SC_{\text{opt}}/N \to \tfrac{872297}{170000} \approx 5.13116}$$

Verification. The values were checked three independent ways, and all three agree exactly: an exact convex min-cost-flow solver on the time-expanded network, the continuous relaxation, and single-agent local search. The potential certificates above were also checked in exact rational arithmetic.